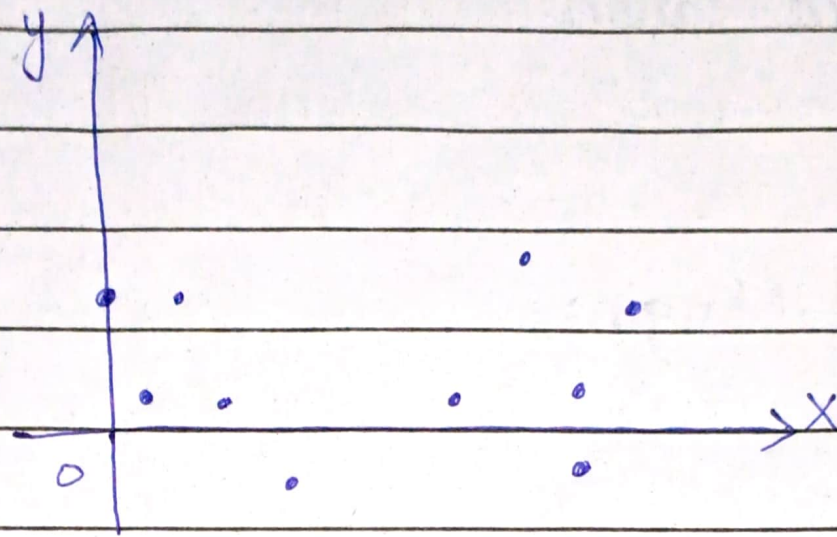
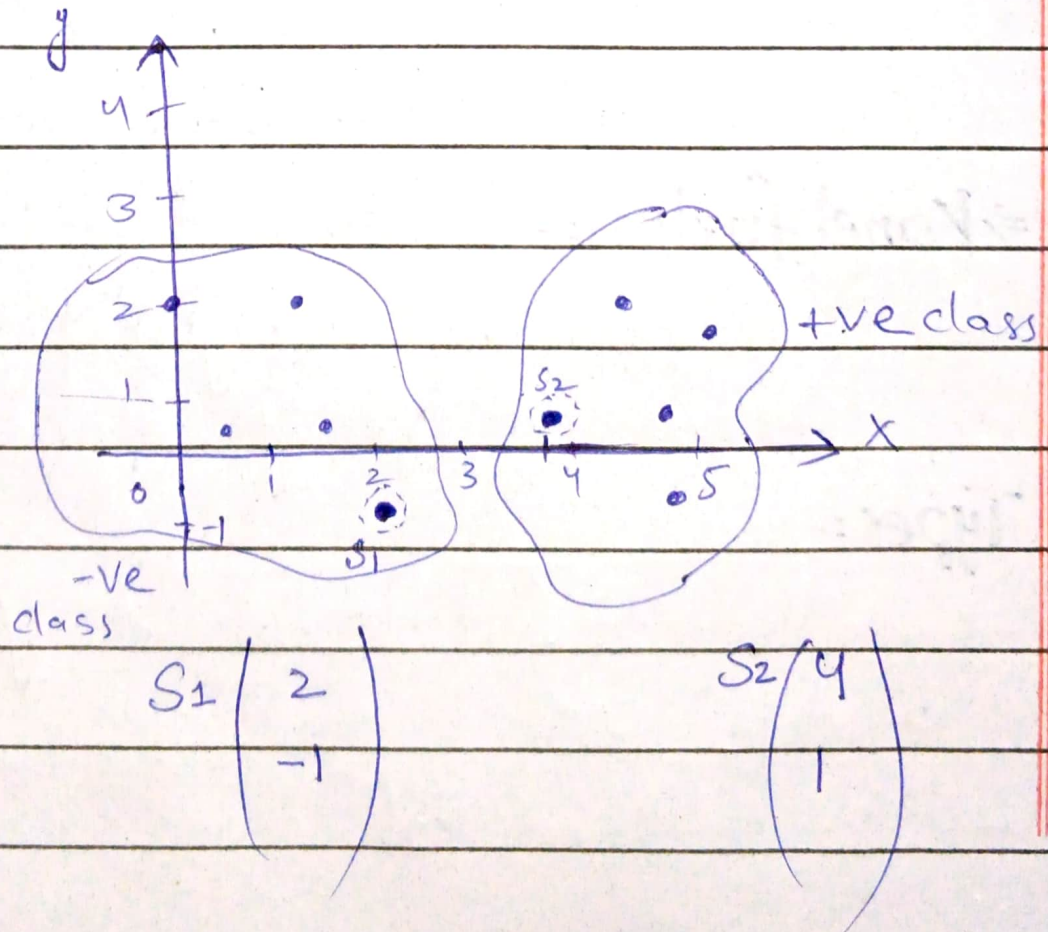


Steps:-

- Draw all the points on plane.



- Find support vectors.



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- Add the bias (b) value in vectors to convert 2D to 3D form.

$$S_1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

α_1

$$S_2 \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}$$

α_2

- Consider Parameters for every support vectors like α_1, α_2 etc.

- Find the linear equations and solve them to find parameter values.

$$\alpha_1 S_1 \cdot S_1 + \alpha_2 S_2 \cdot S_1 = -1 \quad \text{--- (i)}$$

$$\alpha_1 S_1 \cdot S_2 + \alpha_2 S_2 \cdot S_2 = 1 \quad \text{--- (ii)}$$

Put the values

$$\alpha_1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = -1$$

$$\alpha_1 \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 8 \\ -1 \\ 1 \end{pmatrix} = -1$$

$$6\alpha_1 + 8\alpha_2 = -1 \quad \text{--- (iii)}$$

$$\alpha_1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} = 1$$

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$$\alpha_1 \begin{pmatrix} 8 \\ -1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 16 \\ 1 \\ 1 \end{pmatrix} = 1$$

$$8\alpha_1 + 16\alpha_2 = 1 \quad \text{--- (iv)}$$

$$\text{eq (iii)} \times 8 \quad 48\alpha_1 + 64\alpha_2 = -8$$

$$\text{eq (iv)} \times 6 \quad -48\alpha_1 + 108\alpha_2 = -6$$

$$+44\alpha_2 = +14$$

$$\alpha_2 = \frac{14}{44}$$

$$\Rightarrow \boxed{\alpha_2 = 0.318}$$

put α_2 in eq (iii)

$$6\alpha_1 + 8(0.318) = -1$$

$$6\alpha_1 + 2.544 = -1$$

$$\text{--- (i)} \quad 6\alpha_1 = -1 - 2.544$$

$$\text{--- (ii)} \quad 6\alpha_1 = -3.544$$

$$\alpha_1 = \frac{-3.544}{6}$$

$$\Rightarrow \boxed{\alpha_1 = -0.590}$$

- Find the equation for hyperplane is

$$\bar{w} = \sum_{i=1}^n \alpha_i S_i$$

put values

$$\bar{w} = \alpha_1 S_1 + \alpha_2 S_2$$

$$= -0.590 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + 0.318 \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}$$

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$$= \begin{pmatrix} -1.18 \\ 0.590 \\ -0.590 \end{pmatrix} + \begin{pmatrix} 1.272 \\ 0.318 \\ 0.318 \end{pmatrix}$$

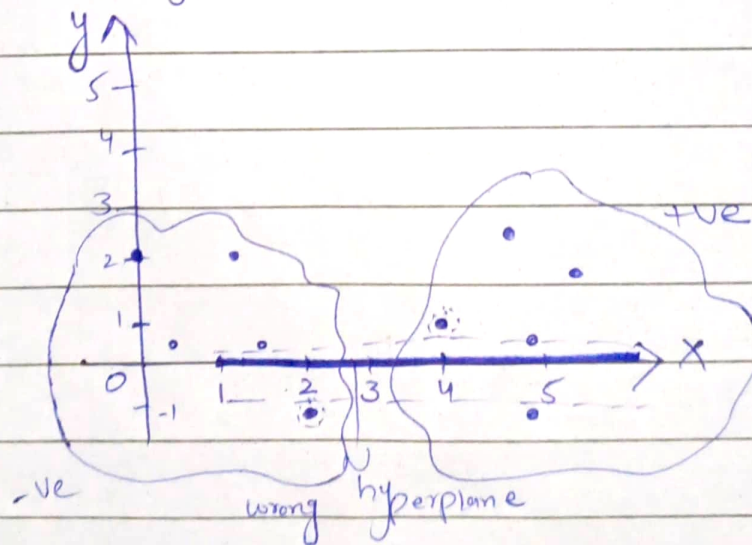
$$= \begin{pmatrix} 0.092 \\ 0.908 \\ -0.272 \end{pmatrix} \approx \begin{pmatrix} 0 \\ 1 \\ -0 \end{pmatrix}$$

Note:-

$$\bar{w} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow \text{orientation of line}$$

bias value $b = 0$

$$\text{line} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \text{vertical}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \text{horizontal}$$

- Draw hyperplane.Note:-Augmented vector method (2D $\rightarrow$ 3D)

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usage, the algorithm minimized the bias(b) value to keep it small, which rotated the hyperplane from a strict vertical line to a horizontal line.